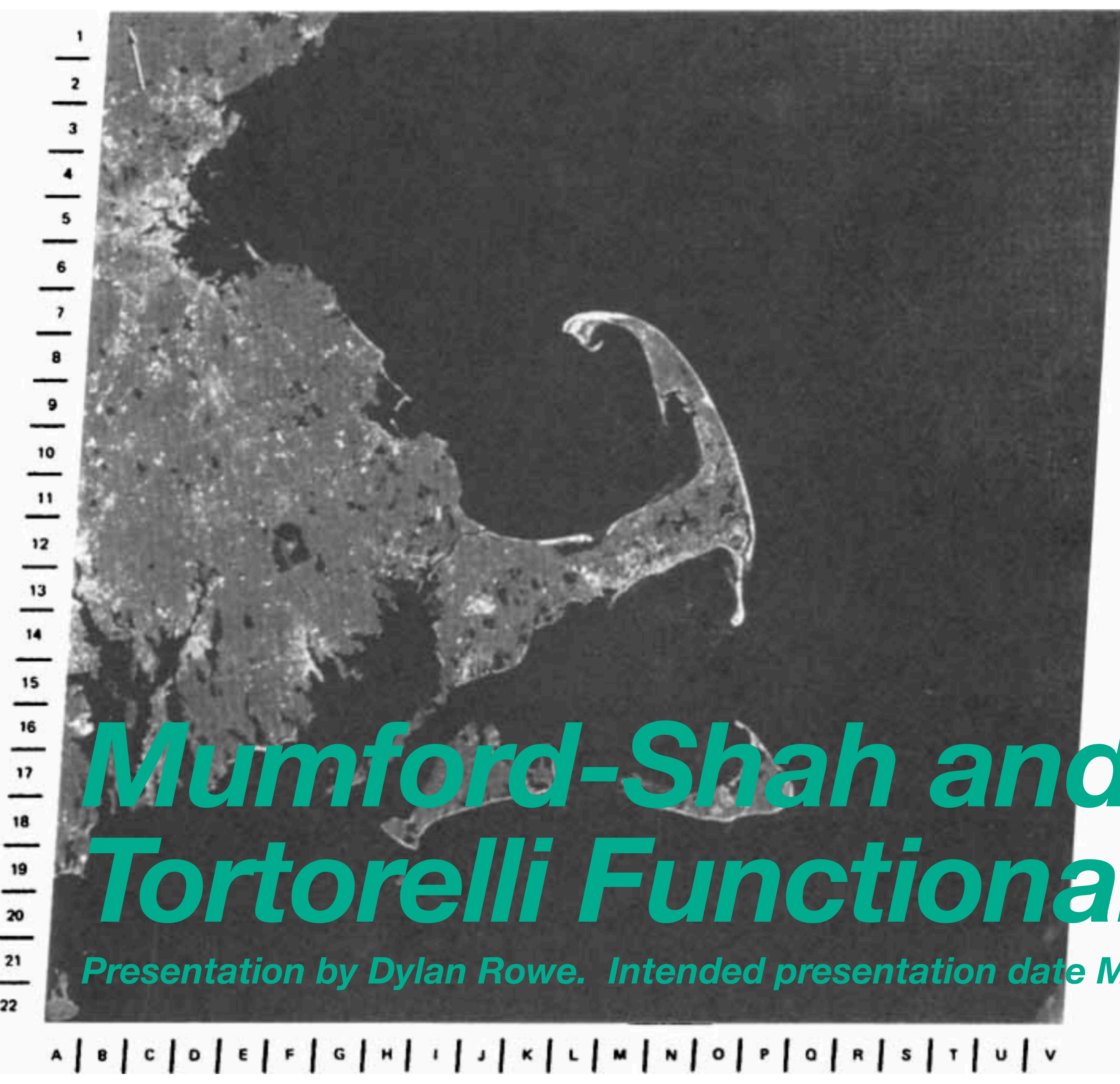




Mumford-Shah and Ambrosio-Tortorelli Functionals on Meshes

Presentation by Dylan Rowe. Intended presentation date March 26th, 2024.

Prerequisites

- The “Dirichlet energy” for a function **penalises variation over some region**

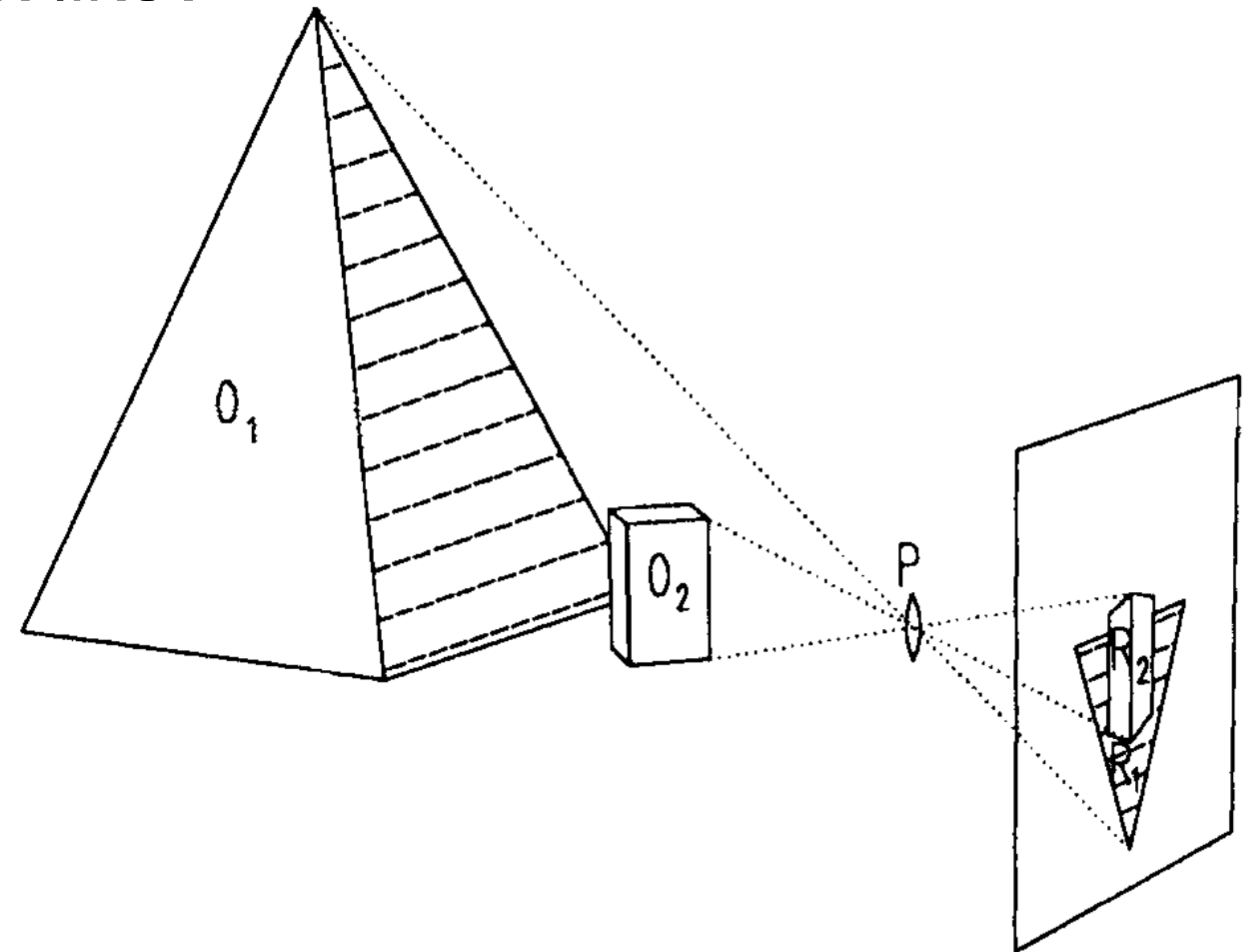
$$E(\boldsymbol{u}) = \int_{\Omega} \|\nabla \boldsymbol{u}\|^2$$

- Similarly, in mesh processing, it describes the **distortion of some map to the plane**
- Minimizers are generally **harmonic functions / maps**

$$\Delta \boldsymbol{u} = 0$$

Mumford-Shah (1989)

- What does the space of natural images look like?
- Without noise:
 - **Smoothly-varying regions**
 - **...with discontinuities between them**



- Image denoising energies should have minimisers matching this description!

Mumford-Shah (1989)

- Given a function u and a curve Γ

$$E(u, \Gamma) = \alpha \int_{\omega} \|u - u_0\|^2 + \int_{\Omega \setminus \Gamma} \|\nabla u\|^2 + \lambda \int_{\Gamma} ds$$

u approximates u_0

u varies little outside of Γ

Γ is short

“ u is an approximation of u_0 which varies little outside of a short Γ ”

Mumford-Shah (1989)

$$E(\boldsymbol{u}, \Gamma) = \alpha \int_{\omega} \|\boldsymbol{u} - u_0\|^2 + \int_{\Omega \setminus \Gamma} \|\nabla \boldsymbol{u}\|^2 + \lambda \int_{\Gamma} ds$$

- Non-convex
 - Difficult to **optimise**
- Lots of research in image processing about making this **reasonable to optimise**

Ambrosio-Tortorelli (1990)

- Invent a function v and a parameter ε (solution is better as $\varepsilon \rightarrow 0$)

$$E(u, \Gamma) = \alpha \int_{\omega} \|u - u_0\|^2 + \int_{\Omega \setminus \Gamma} \|\nabla u\|^2 + \lambda \int_{\Gamma} ds$$

the same

integral is
weighted by v

as $\varepsilon \rightarrow 0$:

- $v = 0$ less frequently
- v gets less smooth

$$E_{AT,\varepsilon}(u, v) = \alpha \int_{\Omega} \|u - u_0\|^2 + \int_{\Omega} \|v \nabla u\|^2 + \lambda \int_{\Omega} \left(\varepsilon \|\nabla v\|^2 + \frac{1}{\varepsilon} \frac{(1 - v)^2}{4} \right)$$

v is an **indicator** for “regions where smoothness (of u) matters”